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White matter changes in reward circuits of amyotrophic lateral sclerosis: a fixel-based study of appetite loss

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Abstract

Background Non-motor symptoms such as appetite loss contribute to weight and fat mass reduction in people living with Amyotrophic Lateral Sclerosis (plwALS), both of which are strong prognostic factors in the disease. Consequently, understanding the neural mechanisms underlying appetite and other non-motor disturbances in ALS is of significant clinical concern. Previous studies highlight widespread grey and white matter involvement beyond the motor system, including hypothalamic volume loss and functional changes in reward-related regions. However, it remains unclear whether structural alterations in white matter tracts implicated in reward processing and behaviour contribute to the multisystem pathology of ALS.

Methods In this case–control study, we employ fixel-based analysis to examine changes in fibre characteristics of non-motor and motor tracts and their associations with clinical, anthropometric, and appetite-related measures within plwALS. Thirty-two plwALS and 24 non-neurodegenerative disease (NND) controls underwent multiband diffusion and structural imaging. Fixel-based analysis was conducted using MRtrix3 to model fibre pathways. For group-level statistical contrasts, fixels were generated in a common template space. We considered case–control differences, appetite, metabolism, body composition, and clinical measures.

Results Results reveal reductions in fibre density and cross-section in the corticospinal/corticobulbar and cerebellothalamic tracts. Exploratory analyses identified fibre density cross-section reductions throughout the temporo-ponto-cerebellar tract, the medial forebrain bundle, and the uncinate fasciculus. Though no direct associations were observed between fibre characteristics and measures of appetite or metabolism, we found significant correlation between fibre cross-section of the corticospinal/corticobulbar tracts and fat-free mass in NND controls, but not in plwALS. Furthermore, disease severity was associated with reduced fibre cross-section in the corticospinal/corticobulbar tract, medial forebrain bundle, uncinate fasciculus, and the temporo-pontine tract.

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Conclusions These findings highlight white matter fibre alterations in both motor and non-motor circuits in ALS. Although direct associations with appetite and metabolism were not observed, the results provide evidence of structural degeneration within reward- and behaviour-related pathways. Taken together, these findings reinforce that ALS is not confined to motor pathways but represents a multisystem neurodegenerative disease with both motor and extra-motor network involvement, offering important insights for future research into disease mechanisms and therapeutic targets.

Keywords Appetite loss, Diffusion MRI, White matter, Fixel-based analysis, Non-motor symptoms

Background

Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disease primarily characterised by the degeneration and death of motor neurons in the brain, brainstem, and spinal cord [1]. Beyond motor involvement, non-motor symptoms such as appetite and metabolic disturbances act as disease modifiers that influence progression and survival [2]. Loss of appetite contributes to malnutrition and subsequent fat mass loss [3]. In turn, fat mass loss is a strong prognostic factor in ALS [4], and so understanding the central mechanisms that shape appetitive behaviours and metabolic regulation is of clinical importance.

The hypothalamus, a key metabolic sensor and appetite regulator of the central nervous system, is widely reported to be impacted in ALS [5]. Hypothalamic pathology has been associated not only with appetite disturbances [6] but also with cognitive changes [7], fat mass loss, and reduced survival [8]. We recently demonstrated differential functional activations of the temporal pole relative to prandial status and of the cerebellum relative to self-reported appetite [3]. These results suggest potential involvement of both temporal and cerebellar regions in disease-related changes. Moreover, imaging studies have identified widespread subcortical and infratentorial grey matter involvement in ALS, including the hypothalamus [8], cerebellum [9], hippocampus, and striatum [10]. Taken together, accumulating evidence indicates that ALS involves focal and widespread alterations across non-motor brain regions, with potential implications for reward, behaviour, and appetite regulation.

A series of interconnected white matter tracts work in concert to create networks that modulate reward and appetitive behaviours. These tracts are diverse, connecting cortical, basal, and infratentorial regions [11]. Diffusion studies in ALS have characterised some of these tracts, but not specifically in relation to appetite and metabolism. Building on evidence of appetite-related regions being affected in ALS, we investigated several key white matter tracts within the appetitive and reward network. These include the uncinate fasciculus, which connects the orbitofrontal cortex with the temporal pole; the medial forebrain bundle, linking the basal forebrain with

the hypothalamus and midbrain [11]; the fornix, connecting the mamillary bodies to the hippocampus [12]; the cingulum bundle, located deep in the cingulate gyrus, with connections ranging from the frontal to the temporal lobe; the hypothalamocerebellar tract, which connects the hypothalamus with the deep cerebellar nuclei; the temporopontine tract, which connects the midbrain with the temporal lobe [13]; and the pontocerebellar tract, which connects midbrain nuclei with the deep cerebellar nuclei [14].

Given that grey matter loss has been observed at the termini of these tracts, it is plausible that tract microstructure is also compromised. To this end, we sought to interrogate these pathways and investigate their associations with anthropometric, appetite, and metabolic measures in people living with ALS (plwALS). As many of these tracts traverse regions with dense crossing fibres—such as the internal capsule and brainstem—we used fixel-based analysis to test for tract-specific fibre alterations. Additionally, we incorporated structural connectome analysis with volumetry to complement tract-specific diffusion measures, providing a comprehensive framework for examining motor and non-motor circuit integrity in ALS, with a focus on appetite-regulating pathways. For completion, we replicate previous work using fixel-based metrics in motor-related tracts [15].

Methods

Participants and study design

Forty-three plwALS were enrolled in this imaging study via Motor Neuron Disease Research clinics at the Wesley Hospital and the Royal Brisbane and Women's Hospital. Inclusion criteria for plwALS included patients who met the diagnostic criteria for possible, probable, or definite ALS [16], no history of gastronomy and/or diabetes, and a forced vital capacity greater than 60% predicted. Twenty-four non-neurodegenerative (NND) controls were recruited as a convenience sample of spouses, friends, or colleagues of plwALS. NND controls had no history of cognitive or neurodegenerative disease. Seven plwALS and two NND controls who did not complete imaging, and four cases who had a final diagnosis

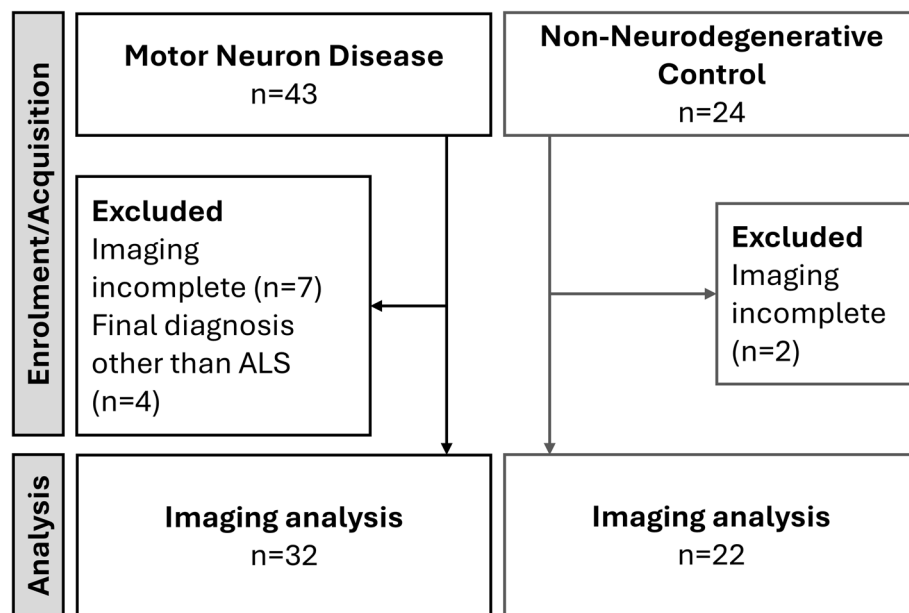


Fig. 1 Participant recruitment, imaging, and analysis of people living with amyotrophic lateral sclerosis (plwALS) and non-neurodegenerative controls

other than ALS, were excluded from downstream imaging analysis. A summary of the participant recruitment, imaging, and analysis is illustrated in Fig. 1.

Metabolic, anthropometric, clinical, and appetite measures were conducted and calculated as previously described [3]. Measurement of subjective appetite were collected using the Council of Nutrition Appetite Questionnaire (CNAQ) [17]. Height was measured using a stadiometer, and body composition was determined using a BodPod system (Cosmed USA Inc., Rome, Italy). Values of fat mass (adipose tissue and stored body fat) and fat-free mass (all nonfat components including muscle) were used to predict resting energy expenditure (REE). Predicted REE was calculated using the Saborini S4 equation [18]. The metabolic index, calculated as the measured resting energy expenditure (Quark RM respirometer, Cosmed) divided by the predicted value, was compared to the NND control cohort. Individuals with a metabolic index greater than one standard deviation above the NND control mean were characterised as being hypermetabolic [19]. The ALS Functional Rating Scale-Revised (ALSFRR-R), a measure of functional decline, was collected from all plwALS at baseline.

Imaging study protocol

We used diffusion-weighted imaging and fixel-based analysis to assess tract-specific fibre properties in ALS and incorporated structural connectome and volumetric analysis for a comprehensive approach. All scans were acquired on a Siemens MAGNETOM Prisma

3 T (Siemens Healthineers, Erlangen, Germany) scanner using a 64-channel head and neck coil. Participants underwent high angular resolution diffusion imaging (HARDI) using a multiband echo planar imaging sequence (multiband factor=2, TR=4700 ms, TE=84 ms, voxel size=2×2×2 mm³, FoV=244×244 mm², acquisition time=2 m:50 s per block, 4 repetitions). The protocol included four blocks collectively comprising 60 unique diffusion directions at b=3000 s/mm², interleaved with b=0 images. Phase encoding directions alternated between anterior-posterior and posterior-anterior across blocks to enable susceptibility-induced distortion correction, performed using FSL's topup algorithm [20]. T1-weighted (T1w) structural scans were obtained in the first session from a three-dimensional 1 mm³ isotropic MP2RAGE sequence (TR=5000 ms, TE=2.98 ms, TI=701 ms, 2500 ms, flip angles=4° and 5°, FoV=256 mm×240 mm, acquisition time=9 m:02 s) [21]. T2-weighted (T2w) fluid-attenuated inversion recovery scans were obtained at 1 mm³ isotropic resolution (TR=5000 ms, TE=386 ms, TI=1800 ms, FoV=256 mm×256 mm, acquisition time=5 m:52 s) [3, 8, 22, 23].

Preprocessing and template construction

Preprocessing, template construction, and all subsequent analyses were conducted in a containerised environment in Neurodesk v1.8.0 [24] and the Bunya supercomputer, The University of Queensland, Australia (<https://doi.org/10.48610/wf6c-qy55>) [25]. Diffusion images were

preprocessed with the Micapipeline pipeline (v0.2.3) for denoising, Gibbs ring correction, distortion and motion correction, and bias field correction [26]. Images were then processed in MRTrix (v3.0.4) for multi-tissue constrained spherical deconvolution to calculate group-averaged fibre orientation distribution (FODs) [27]. The average tissue response functions were incorporated as part of multi-tissue spherical deconvolution to estimate the fibre orientation distribution function (fODF) [27]. FODs within a brain mask generated by SynthStrip (v1.7) [28] were normalised using multi-tissue informed log-domain intensity normalisation [29]. T1w and T2w images were corrected for N4 field bias and denoised using Advanced Normalisation Tools (ANTs, v2.5.2) [30].

Group FOD templates were initially generated for plwALS and controls separately. We used a multimodal approach in ANTs (v2.5.2) [30] to generate group-wise tissue-unbiased templates by incorporating anatomical (T1w and T2w) and diffusion volumes (apparent diffusion coefficient and fractional anisotropy) as inputs [31]. A mid-space common template was generated by combining the two cohort templates using multimodal non-linear registration. Subject-specific FODs were transformed to the common template space using the generated transform files. Subsequently, FODs were combined to create a study-wide FOD template in MRTrix [32]. Similarly, a study-wide T1w template was generated in ANTs on the transformed T1w images.

Fixel-based analysis

Fixel fibre density and cross-section metrics for each participant were generated with MRTrix [33]. Next, tissue segmentation and grey-white matter boundaries were generated using the “fsl” algorithm on the T1w template [34]. This was used to seed the anatomically constrained tractography to generate 40 million streamlines from the FOD template. The iFOD2 algorithm was used with the default step size of half a voxel, maximum angle of 45°, minimum length of 5, maximum length of 250, and a cut-off amplitude of 0.06 [35, 36]. Backtracking was enabled to allow for the truncation of tracts and retracking where suboptimal streamline terminations occur. The SIFT algorithm was used to filter tractograms down to four million streamlines [37].

We limited the fixel-based analysis to specific tracts with relevance to appetite and metabolism. These include the corticospinal/corticobulbar tracts, cerebellothalamic tract (with a focus on primary motor connections), medial forebrain bundle, uncinate fasciculus, cingulum bundle, fornix, pontocerebellar tract, temporopontine tract, and hypothalamocerebellar tract (Fig. 2). We previously identified potential involvement of the hypothalamus [8], cerebellum, and temporal pole [3] in appetite regulation in

ALS. Thus, while not traditionally considered part of the appetite reward system, inclusion of the pontocerebellar, temporopontine, and hypothalamocerebellar tracts allowed for testing of the hypothesis that the cerebellum and cerebro-cerebellar connectivity are involved in the disease and may contribute to behavioural changes [9], such as the loss of appetite.

Tracts of interest were extracted using rules-based selection of streamlines using a combination of terminating and projecting structures and tract bodies. Segmentations of structures were defined from various atlases or segmentation tools described below. The temporal lobe, orbitofrontal gyrus, precentral gyrus, and thalamus were derived from the Hammer atlas [38]. The hypothalamus was extracted using OSHy-X [39]. The midbrain and pons were derived from a Bayesian implementation of brainstem structure segmentation [40]. The temporal pole and medial temporal lobe were derived from FreeSurfer (v7.4.1) [41]. The cerebellum was derived from CerebNet (FastSurfer v2.4.2) [42]. Masks for the bodies of the fornix and cingulum were derived from the JHU atlas [43].

The corticospinal/corticobulbar tract was generated by selecting streamlines that traverse the precentral gyrus, internal capsule, midbrain, and pons while excluding streamlines that also traverse the cerebellum and thalamus. The cerebellothalamic tract was generated by selecting streamlines that traverse the cerebellum, thalamus, and precentral gyrus. The medial forebrain bundle was generated by selecting streamlines that traverse the midbrain and the medial orbitofrontal cortex. The uncinate fasciculus was generated by selecting streamlines that traverse the orbitofrontal cortex and the temporal pole. The cingulum bundle was generated by selecting streamlines that traverse the JHU atlas cingulum mask. The fornix was generated by selecting streamlines that traverse the JHU atlas fornix body mask. The pontocerebellar tract was generated by selecting streamlines that traverse the pons and cerebellum. The temporopontine tract was generated by selecting streamlines that traverse the temporal lobe and pons. Left and right components were generated separately for bilateral tracts. Tracts were inspected manually (by J. C. and H. L.) and corrected for erroneous streamlines. The code used to generate these tracts are available on OpenNeuro: <https://doi.org/10.18112/openneuro.ds005874.v1.1.0> [22, 23]. A 3D video visualisation of these tracts and the masks are shown in Supplementary Video 1: <https://doi.org/10.6084/m9.figshare.30511307> [44].

Fibre cross-section, fibre density, and fibre density-cross-section were tested in a fixel-wise general linear model with 5000 permutations [45]. A connectivity-based fixel enhancement approach was used to identify structurally connected significant fixels [45]. For the

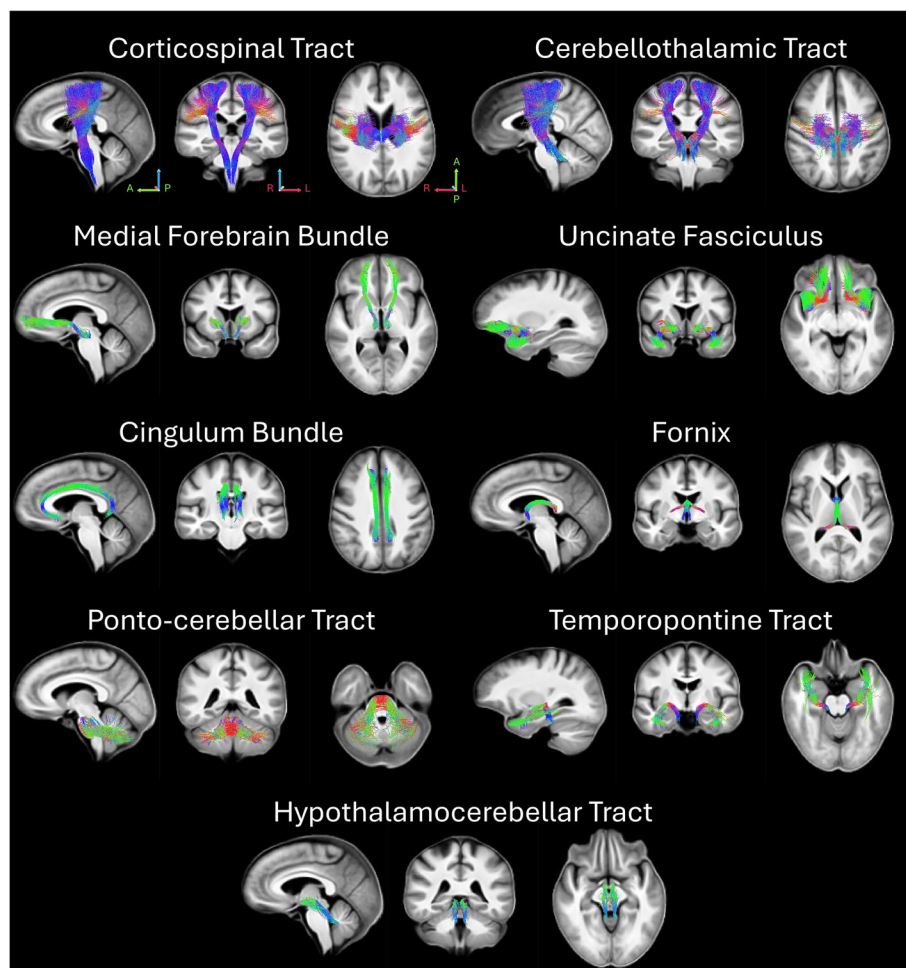


Fig. 2 Tractography of nine white matter tracts of interest. Tracts are coloured by direction: Blue indicates superior-inferior, green indicates anterior-posterior, and red indicates left-right direction. A, anterior; P, posterior; R, right; L, left

corticospinal/corticobulbar and cerebellothalamic tracts, we considered the statistical contrast, Control > plwALS, accounting for age and sex in a general linear model. Fixels were considered significant if the family-wise error-corrected p -value was less than 0.05. Family-wise error correction was conducted using permutation testing with 5000 permutations [46].

We next considered the non-motor tracts together. Given the exploratory nature of this study, both corrected and uncorrected findings were examined to identify potential tracts of interest for further validation. From the Control > plwALS contrast, we identified the medial forebrain bundle, uncinate fasciculus, temporopontine, and ponto-cerebellar tracts for further analysis (Fig. 2). We analysed the fibre properties within these tracts in relation to appetite scores (CNAQ) [17], metabolic index, Δ FRS (used here as a measure of disease progression rate, calculated as the difference between the maximum ALSFRS-R score [48] and the participant's score at the

time of imaging, normalised by the months elapsed since symptom onset) [47], fat-free and fat mass at the time of imaging in plwALS, and adjusting for age and sex. Analysis of fat-free mass included height as an additional nuisance variable. Where uncorrected significant fixels were identified, fibre values were extracted from these fixels, and a correlation analysis was conducted with the clinical, appetite, or metabolic measure.

Visualisation of fixel results was achieved in MRTrix [33] by overlaying thresholded fixels over individually coloured tracts. The fixel length multiplier was set to 1.5. The main image was set to an inverted and scaled study-wide brain mask.

Structural connectome

To gain a system's understanding of the appetite and behaviour networks from the fixel-based analysis, we conducted a structural connectome analysis [48]. Streamlines for each participant were generated using

Table 1 Baseline characteristics of people living with amyotrophic lateral sclerosis (plwALS) and non-neurodegenerative controls (NND controls) at time of baseline assessment

	Amyotrophic lateral sclerosis (N= 32)	Non-neurodegenerative controls (N= 22)	p-value
Demographic, anthropometric, and appetite measures			
Age	58.72 (8.93)	56.32 (10.51)	0.37
Sex (females)	8 (0.25)	6 (0.27)	1.00
CNAQ total	29.66 (2.78)	31.64 (2.94)	0.02
Weight (kg)	82.43 (15.93)	85.09 (20.79)	0.60
BMI	27.55 (5.01)	27.55 (6.56)	1.00
FM (kg)	29.02 (14.41)	27.08 (17.52)	0.66
FFM (kg)	53.41 (9.66)	58.03 (10.1)	0.10
Resting energy expenditure (kcal/day)	1721.97 (370.72)	1727.73 (356.72)	0.81
Metabolic index	1.08 (0.21)	1 (0.11)	0.10
Hypermetabolic	13 (0.41)	5 (0.23)	0.24
Clinical measures			
ALSFRS-R total	38.58 (5.45)		
Diagnostic delay (months)	12.97 (7.98)		
Time since onset (months)	18.13 (6.30)		
ΔFRS	0.55 (0.34)		
Site of onset (bulbar)	5 (0.16)		

ALS amyotrophic lateral sclerosis, ALSFRS-R ALS functional rating scale-revised, BMI body mass index, CNAQ Council on Nutrition Appetite Questionnaire, ΔFRS decline in the ALSFRS-R (points/month since date of symptom onset). Data presented as mean (standard deviation) for continuous variables or *n* counts (%) for categorical variables. Student two-tailed *t*-test with Welch’s correction is conducted for continuous variables. Fisher’s exact test is conducted for categorical variables

anatomically constrained tractography and filtered using the SIFT2 algorithm, a supplement to the original SIFT algorithm that enables quantitative connectome analyses [49]. Key structures used to select the tracts for the fixel-based analysis were selected to represent nodes in the connectome. This includes the midbrain, pons, and the left and right hemispheres of the cerebellum, hypothalamus, medial temporal lobe, medial and lateral orbitofrontal gyrus, superior temporal gyrus, and the temporal pole. Masks for these structures were derived from sources as previously described under “Fixel-based analysis”. Connectomes were generated by MRTrx with contributions to connectome edges scaled by the inverse of the node volumes of each pairwise connection [50]. A control > plwALS contrast was conducted using the threshold-free network-based statistics algorithm [51]. Age and sex were included as covariates in the design matrix, and edges with family-wise error-corrected *p*-values less than 0.05 were considered significant. Family-wise error correction was conducted using permutation testing with 5000 permutations [46]. Visualisation of connectome results was achieved using Nichord (v 0.3.3) [52].

Volumetric analysis

Two-tailed Student *t*-tests were conducted to test for differences in means between plwALS and NND controls. Structures that were tested include the midbrain, pons,

cerebellum cortex, thalamus, medial temporal lobe, primary motor cortex, superior temporal gyrus, temporal pole, and orbitofrontal gyrus. Volumetric measures were adjusted by total intracranial volume, and *p*-values were adjusted for multiple testing using the false discovery rate (FDR). *p*-values less than 0.05 were considered significant.

Results

Participant demographics, anthropometric, metabolic, appetite, and clinical measures

Thirty-two plwALS and 22 NND controls underwent imaging and baseline assessments (Fig. 1). Demographics, anthropometry, and metabolic measures were not significantly different between cohorts (Table 1), including age (*p* = 0.37), sex (*p* = 1.00), weight (*p* = 0.60), body mass index (BMI, *p* = 1.00), fat mass (*p* = 0.66), fat-free mass (*p* = 0.10), resting energy expenditure (*p* = 0.81), metabolic index (*p* = 0.10), and hypermetabolism (*p* = 0.24). Consistent with prior reports [53], subjective appetite as measured by the CNAQ was significantly lower in plwALS compared with controls (*p* = 0.02). CNAQ scores did not correlate with the ALSFRS-R score (*p* = 0.327).

Case-control fixel-based analysis

Fixel-based analysis reveals significant differences in fibre density and cross-section between plwALS and NND

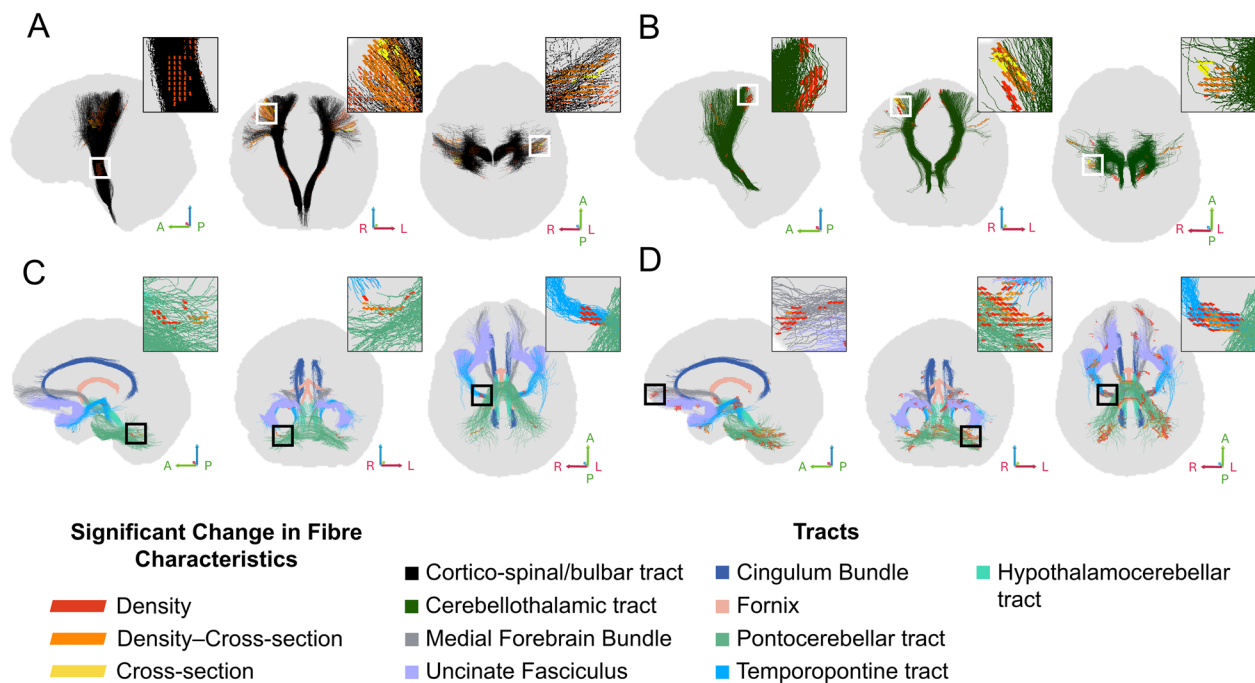


Fig. 3 Significant change in fibre characteristics between non-neurodegenerative controls ($N=22$) and people living with amyotrophic lateral sclerosis (plwALS, $N=32$) in the **A** corticospinal/corticobulbar tracts, **B** cerebellothalamic tracts and appetite and behaviour-related tracts, and **C** family-wise error corrected and **D** non-corrected. A, anterior; P, posterior; R, right; L, left

controls in the motor tracts. Fibre density and cross-section are significantly reduced in plwALS throughout the corona radiata of the precentral gyrus in the corticospinal/corticobulbar tracts (Fig. 3A) and the cerebellothalamic tract (Fig. 3B). Pure fibre density reductions are evident in the internal capsule region of the corticospinal/corticobulbar tract and to a lesser extent in the cerebellothalamic tract.

We next considered case–control differences in the non-motor tracts, namely the medial forebrain bundle, uncinate fasciculus, cingulum bundle, fornix, pontocerebellar tract, temporopontine tract, and the hypothalamocerebellar tracts. Family-wise error-corrected results show sparse fibre density and cross-section loss in the bilateral cerebellum (Fig. 3C, coronal view). Additionally, the bilateral sublenticular segment of the internal capsule, which forms part of the temporopontine connection, shows significant pure fibre density loss (Fig. 3C, axial view inset). We next examined the uncorrected results (Fig. 3D). We observe broader fibre density and cross-sectional reductions throughout the cerebellum, the sublenticular segment of the internal capsule, and the anterior portions of the temporopontine tract. Clusters of predominantly fibre density reductions are observed throughout the medial forebrain bundle and the uncinate fasciculus. Some isolated fixels were detected in the cingulum bundle, fornix, and

the hypothalamocerebellar tracts, suggesting possible involvement of these pathways in ALS.

Clinical, appetite, and metabolic measures

A fixel-based analysis was conducted within each cohort to investigate the associations between fibre characteristics in motor and non-motor tracts with clinical, appetite, and metabolic measures. In the NND control group, uncorrected analyses showed that fat-free mass at the time of imaging was associated with reduced fibre cross-section in the corticospinal tract (Fig. 4A). This relationship was supported by a positive correlation between extracted fibre cross-section fixels and fat-free mass ($\rho=0.71$, $p<0.001$). In contrast, when including an interaction term in the model to test for an association in plwALS, we found a potential attenuation of this relationship (Fig. 4B). The result suggests that these differences in the relationship of fibre cross-section with fat-free mass occur throughout the corticospinal/corticobulbar tracts. Fibre cross-section values from the identified fixels were extracted and incorporated into a linear model examining the effect of fat-free mass and cohort on the averaged cross-section values of extracted fixels [$F(3,49)=14.03$, $p<0.001$; $R^2=0.462$]. The model revealed a main effect of fat-free mass ($\beta=7.71\times 10^{-6}$, $p=0.017$) and cohort ($\beta=7.52\times 10^{-4}$, $p=0.008$) on fibre cross-section. An interaction was observed between fat-free mass and

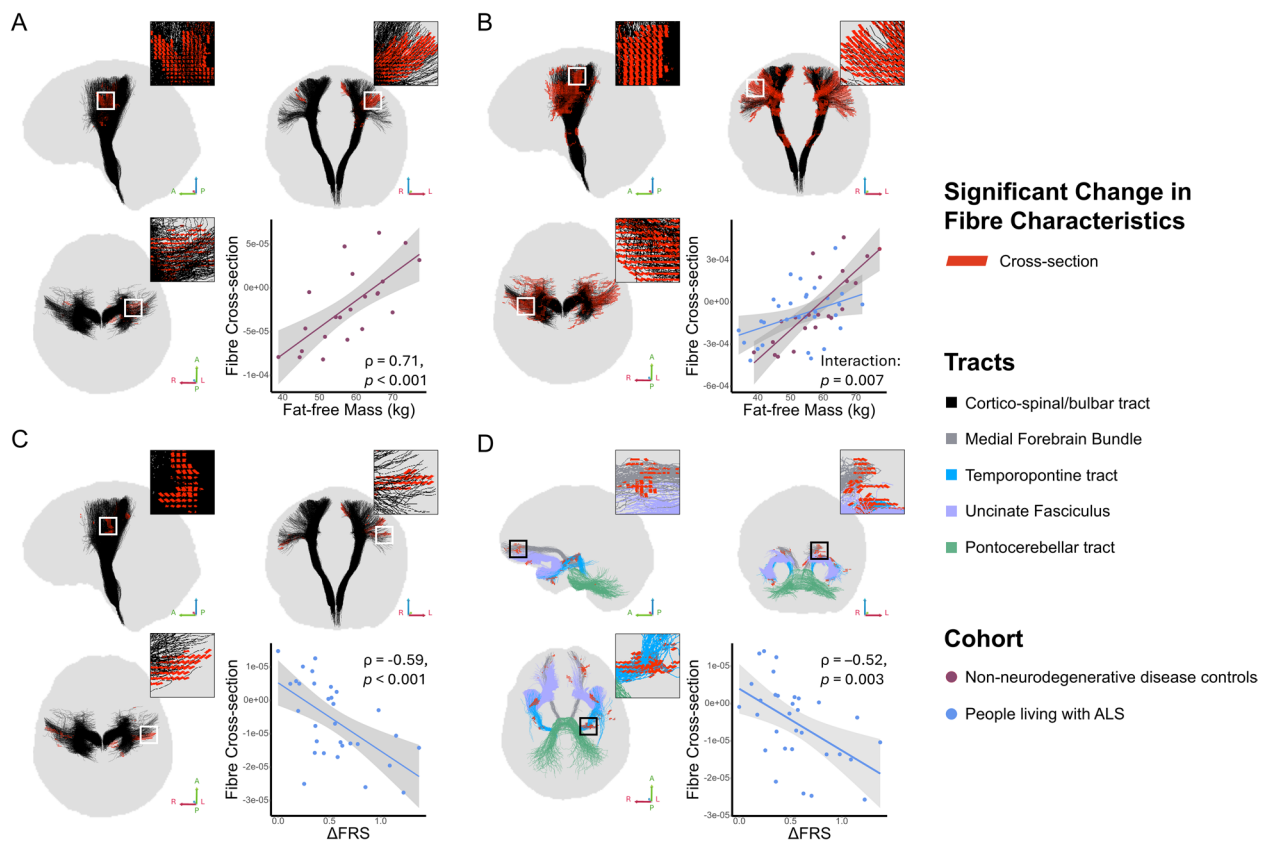


Fig. 4 Association (uncorrected) of corticospinal/corticobulbar fibre cross-section relative to baseline fat-free mass (**A**) in non-neurodegenerative disease controls ($N=22$) and **B** interaction with cohort. Association of fibre cross-section relative to Δ FRS in people living with amyotrophic lateral sclerosis (plwALS, $N=32$) in **C** corticospinal/corticobulbar tracts and **D** appetite and behaviour-related tracts of interest. Scatterplots show each participant's average fixel cross-section values against the measure of interest. Fibre cross-section values have been transformed using the natural logarithm. A, anterior; P, posterior; R, right; L, left

cohort ($\beta = 1.34 \times 10^{-5}$, $p = 0.007$), suggesting that the relationship between fat-free mass and fibre cross-section was preserved in controls but attenuated in plwALS.

Next, uncorrected analyses for Δ FRS in plwALS indicated an association with fixel cross-section in the corona radiata of the corticospinal/corticobulbar tracts (Fig. 4C). This relationship was supported by a negative correlation between extracted fibre cross-section fixels and Δ FRS ($\rho = -0.59$, $p < 0.001$). Additionally, when considering non-motor tracts of interest identified in the case-control fixel-based analysis, results indicate an association of Δ FRS with fixel cross-section in the medial forebrain bundle and temporopontine tract (Fig. 4D). Correlation analysis similarly supported this result, indicating a negative correlation between extracted fibre cross-section fixels and Δ FRS ($\rho = -0.52$, $p = 0.003$).

No associations of clinical, appetite, and metabolic measures were detected for fibre density and fibre density-cross-section measures.

Structural connectome

A structural connectome analysis [48] was conducted to map and test the connectivity (streamlines) of 16 structures or nodes throughout the non-motor tracts of interest, hypothesised to be involved in the loss of appetite in plwALS. This analysis tests the strength of connections on the network level which complements the results from the fixel-based analysis. FDR-corrected significant connections are reported in Table 2. The top 7 connections (ordered by p -value) are visualised on a glass brain in Fig. 5A, and all connections are visualised in a chord diagram in Fig. 5B. Results suggest that in plwALS, significant reductions in connectivity occur between the midbrain and the left ($p < 0.001$) and right ($p < 0.001$) medial temporal lobe, suggesting temporopontine tract involvement; between the pons and the left medial orbitofrontal gyrus ($p = 0.011$), suggesting medial forebrain bundle involvement; and between the midbrain and pons ($p = 0.019$), suggesting brainstem involvement. Additional connectome statistics are provided in Additional file 1: Table S1.

Table 2 Significant connections of the structural connectome between amyotrophic lateral sclerosis ($N=32$) and non-neurodegenerative controls ($N=22$)

Significant connections	Tract system involvement	p -value (FWE)
Midbrain $\longleftrightarrow$ left medial temporal lobe	Temporopontine tract	<0.001
Midbrain $\longleftrightarrow$ right medial temporal lobe	Temporopontine tract	<0.001
Pons $\longleftrightarrow$ Left medial orbitofrontal gyrus	Medial forebrain bundle	0.011
Midbrain $\longleftrightarrow$ pons	Brainstem	0.019

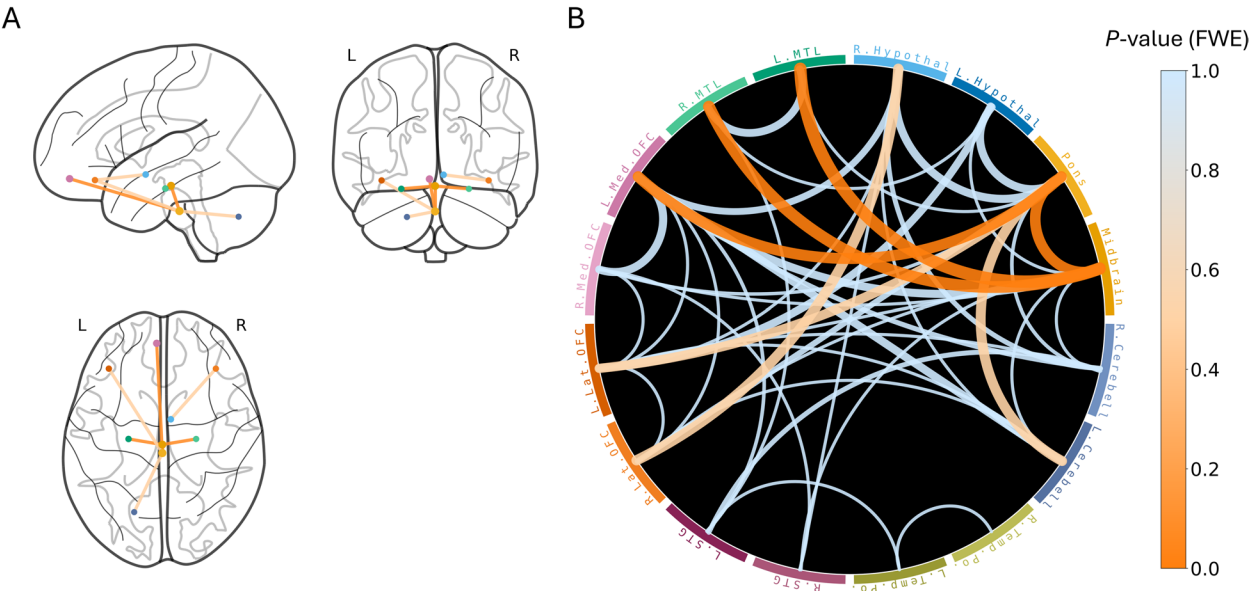


Fig. 5 Differences in structural connectome results of non-motor regions between amyotrophic lateral sclerosis ($N=32$) and non-neurodegenerative controls ($N=22$). **A** The top 7 connections by p -value are visualised on a glass brain. Edge colours correspond to family-wise error-corrected p -values, and dots represent nodes coloured by the structures they represent, as indicated in **B**. **B** Full connectome results visualised in a chord diagram. Each chord represents a connection between nodes, their colours represent family-wise error-corrected p -values, and chord thickness is scaled by effect size. MTL, medial temporal lobe; Hypothal, hypothalamus; Cerebell, cerebellum; Temp Po, temporal pole; STG, superior temporal gyrus; Lat OFC, lateral orbitofrontal cortex; Med, medial orbitofrontal cortex

Volumetric and cortical thickness measures

We sought to test for group differences in volumetry and cortical thickness of key projecting or terminating structures of identified tracts of interest (Table 3). FDR-adjusted comparisons suggest significant cortical thickness reductions in the primary motor cortex ($p=0.020$), superior temporal gyrus ($p=0.048$), and orbitofrontal gyrus ($p=0.049$). No cortical thickness differences were observed for the temporal pole ($p=0.107$), and no significant volumetric differences were observed for the midbrain ($p=0.950$), pons ($p=0.935$), cerebellum cortex ($p=0.935$), medial temporal lobe ($p=0.935$), and the thalamus ($p=0.935$). When considering bilateral structures, a reduction in average thickness of the left and right primary motor cortex trended towards significance (FDR -adjusted $p=0.065$, Additional file 2: Table S2).

Discussion

This study aimed to assess microstructural alterations in appetite and behaviour tracts in plwALS. Non-motor brain deficits are commonly reported in ALS [10], including the hypothalamus [8]. These changes are often reported relative to non-motor cognitive and behavioural impairments [10], which includes changes in appetitive behaviours. Collectively, our findings suggest reductions in fibre density, cross-section, and connectivity across several non-motor tracts; while no associations were observed with appetite and metabolic measure, these results point to a wider involvement in ALS, including networks that may contribute to appetitive behaviours. Results of fibre density and cross-section deficits in the corticospinal/corticobulbar tracts and cerebellothalamic tract in plwALS confirm a well-established consensus for

Table 3 Comparison of volumetric and cortical thickness measures between amyotrophic lateral sclerosis ($N = 32$) and non-neurodegenerative controls ($N = 22$)

	Tract involvement	ALS ($N = 32$)	Control ($N = 22$)	p -value (uncorrected)	p -value (FDR)
Volumes (mm^3)					
Midbrain	Medial forebrain bundle	6244.19 (400.07)	6236.55 (491.74)	0.950	0.950
Pons	Temporopontine, pontocerebellar	15,322.64 (1261.34)	15,607.17 (1774.89)	0.494	0.935
Cerebellum cortex	Pontocerebellar	117,788 (9237.48)	116,806.3 (13,089.49)	0.748	0.935
Thalamus	Cerebellothalamic	1349.1 (1184.11)	13,880 (1127.38)	0.232	0.935
Medial temporal lobe	Fornix, temporopontine	7826.038 (825.726)	7923.97 (679.21)	0.648	0.935
Average thickness (mm)					
Primary motor cortex (precentral + paracentral)	Corticospinal/corticobulbar tracts	2.14 (0.17)	2.27 (0.16)	0.005	0.020
Superior temporal gyrus	Temporopontine	2.61 (0.17)	2.72 (0.16)	0.024	0.048
Temporal pole	Uncinate fasciculus	3.18 (0.28)	3.32 (0.34)	0.107	0.107
Orbitofrontal gyrus (medial + lateral)	Medial forebrain bundle, uncinate fasciculus	2.40 (0.17)	2.50 (0.19)	0.037	0.049

microstructural alterations in motor pathways, a consequence of upper motor neuron degeneration/death in ALS. Specifically, alterations in the internal capsule and corona radiata are consistent with results from prior literature using fixel-based analysis [54], neurite orientation and dispersion density imaging (NODDI) [55, 56], and diffusion tensor imaging [57]. This is complemented by the reductions in the primary motor cortex in this cohort, suggesting neuronal degeneration and death in the motor tracts.

In NND controls, the association between fat-free mass and corticospinal tract cross-section may reflect general interindividual variation in motor system organisation and muscle function. This relationship is particularly relevant because skeletal muscle mass constitutes a major component of fat-free mass. The apparent weakening of this association in plwALS is consistent with neurodegenerative processes disrupting the normal coupling between motor pathway tract integrity and muscle status. This attenuation suggests that ALS-related degeneration could interfere with coupling mechanisms that typically link motor tract integrity and muscle function observed in controls. Disruption of neuroimaging associations by underlying pathology has been reported in other disorders; for example, motor symptoms and age contribute to reduced coupling between cortical curvature and thickness in Parkinson's disease [58], and the relationship between dentate-nuclei volume and susceptibility is observed in healthy controls but not in individuals with Friedreich's ataxia [59].

Our interrogation of non-motor tracts suggests fibre density and cross-sectional changes in plwALS. Notably, fibre alterations were detected in tracts with direct

connections to the brainstem, including the pontocerebellar and temporopontine tracts. The brainstem is a critical hub for regulating basic physiological functions such as breathing, heart rate, motor, and sensory functions. It is also a crucial thoroughfare for projection tracts like the corticospinal tract. While we did not observe alterations in volumes within the brainstem, reductions in connectome connectivity were observed between the pons and midbrain in plwALS. This suggests that there is matter involvement and a decrease in network strength within these brainstem structure, even when gross anatomical changes are not yet evident. At the time of assessment, participants had a mean ALS-FRS-R score of 38 at approximately 18 months from symptom onset (Table 1). Although grey matter brain volumes did not differ between groups, the presence of white matter microstructural changes suggests that neurodegenerative processes were already underway before overt volumetric loss becomes apparent. Such microscopic changes likely accumulate over time and can precede macroscopic alterations detectable using conventional structural imaging techniques and non-specific segmentation software. Findings underscore the importance of sensitive imaging techniques for identifying microstructural changes [15, 60] that might occur prior to gross volumetric changes on routine T1w magnetic resonance images.

The brainstem is recognised as a primary and secondary site of pathology in ALS [61]. Pathological spread is a common hypothesis in ALS, backed by evidence of focal onset followed by the spread of symptoms as the disease progresses [62]. On the cellular level, neurons with mislocalised and misfolded proteins have a higher

propensity for degeneration. It is hypothesised that this is accompanied by the seeding and cell-to-cell propagation of misfolded proteins, leading to the regional spread of pathology [63]. This cascade of grey matter neuronal death can, in turn, influence the degeneration of brainstem-connected white matter tracts, compromising their structural integrity and fibre characteristics. True evidence of the spread of white matter degeneration, however, will require longitudinal imaging studies of clinically well-characterised cohorts. Postmortem studies in ALS provide convergent evidence supporting these mechanisms. Phosphorylated TDP-43 (pTDP-43) pathology accumulates in neurons and glia and spreads along axonal pathways [61], providing a biological basis for white-matter microstructural change. Focal onset and contiguous propagation across motor systems are well documented in ALS [64], consistent with tract-based vulnerability. Oligodendrocyte and myelin pathology within ALS white matter are also substantial and may precede visible macrostructural loss [65], in line with microstructural alterations observed here. At a systems level, both brainstem and cerebellar pathology have been reported [66], supporting extra-motor circuits relevant to our findings.

The neuropathological observations noted above are consistent with neuroimaging evidence demonstrating that ALS pathology extends beyond cortical motor regions to subcortical and infratentorial regions, including the cerebellum. Evidence supporting this includes observed reductions in cerebellar volume and alterations in diffusion tensor imaging metrics within the cerebellum [9]. Additionally, changes have been identified in cortical connections that project to the brainstem and cerebellum [9]. Pontocerebellar neurons project from the pontine nuclei and terminate in the cerebellar cortex, a tract recognised to be involved in numerous functions including motor control and cognitive processing [67, 68]. While we did not observe volumetric changes in the pons or cerebellar cortex, we observed reductions in fibre density and cross-section in the cerebellar white matter of the pontocerebellar tract in plwALS, indicating potential degeneration of axons but not cell bodies. The involvement of the cerebellum in non-motor roles like reward and appetite is emerging. Cerebellar activity has previously been shown to be modulated by food cues in healthy controls [69] and plwALS [3]. In anorexia and bulimia nervosa, lobule VI of the cerebellum is hypothesised to be involved in altered eating behaviours through sensorimotor and emotional learning mechanisms, while in obesity Crus I is hypothesised to be involved through executive function mechanisms [70]—lobules that are consistent with the significant fixel results (corrected) presented in Fig. 3C. Importantly, we did not observe

associations of fibre characteristics with appetite scores in the cerebellum, suggesting a potentially more complex contribution of the structure to appetite. Nevertheless, other aspects of reward, emotion, and executive function will be features of interest for future studies in the cerebellum in ALS.

Upstream, we report potential reductions of fibre density and cross-section of the temporo-ponto connection in the sublenticular segment of the internal capsule and reduced cortical thickness in the superior temporal gyrus in plwALS. However, we note that the proximity of the sublenticular segment to the corticospinal tract may be subject to registration accuracy and partial volume effects. Temporopontine fibres originate from various regions of the temporal lobe [67], including the superior temporal gyrus, which was first identified as a projection site for temporopontine fibres [71]. The fibres collectively converge and descend through the sublenticular segment of the internal capsule, where they terminate in the pontine nuclei [72]. The fixel-based results for this tract are supported by our structural connectome findings that suggest a disruption or attenuation of the structural connection strength between the temporal lobe and brainstem. Together, results provide additional evidence for focal involvement of fibres and degradation of connection strength in the temporopontine tract. However, more comprehensive studies are necessary to confirm these exploratory findings and understand how degeneration of this tract contributes to non-motor symptoms in ALS.

The midbrain houses reward-modulating nuclei, notably the ventral tegmental area and the substantia nigra pars compacta. These structures contain dense collections of dopaminergic neurons. The medial forebrain bundle is a complex bidirectional pathway that courses through the lateral hypothalamus and consists of fibres connecting the midbrain nuclei with basal forebrain structures, including the orbitofrontal cortex [11]. This tract facilitates reward-related information processing and decision-making. The orbitofrontal cortex is also connected to the temporal pole through a major association tract, the uncinate fasciculus [11]. Observations of reduced connectome connectivity and possible fibre property alterations in the medial forebrain bundle and uncinate fasciculus, together with reduced cortical thickness in the orbitofrontal gyrus, suggest that connectivity and the central regulation of reward in this region may be impacted. However, we note that appetite scores were not associated with fibre properties in these tracts. Instead, uncorrected results suggest that like the corticospinal/corticobulbar tracts, Δ FRS may be negatively correlated with the cross-section of some fibres in the medial forebrain bundle, uncinate fasciculus, and temporopontine

tract. Residing in the frontal and temporal lobes, degradation of these tracts could represent broader cognitive and behavioural decline. These lobes are also the primary sites of pathology for frontotemporal dementia, a disease that clinically sits on a continuum with ALS [73]. The physical connections and potential pathological spread between these primary sites of pathology could therefore be an area of interest in future investigations of the ALS-FTD spectrum.

The absence of changes in fibre property with appetite may be indicative of the complex, multifactorial nature of non-motor features in ALS. This null result is instructive. Our results to date suggest that plwALS have altered functional responses to food in the temporal pole and cerebellum [3]. Structural alterations of appetite and behaviour tracts alone do not explain the loss of appetite in ALS. Instead, there may be a variety of other factors that contribute to the complex modulatory deficits for appetite and metabolism in plwALS, including neuroendocrine [74], peripheral [75], and psychosocial factors [76]. Additionally, more distributed neuronal changes throughout the brain with distinct focal white-matter-specific changes may contribute to functional changes that alter appetite. Imaging studies of plwALS in the last few decades have demonstrated broad grey matter alterations of subcortical structures related to behavioural changes, including the hypothalamus [7], hippocampus [77], and striatum [78]. Importantly, results demonstrate that the non-motor clinical features in ALS, such as appetite and metabolism, may not simply reflect the degree of white matter structural degradation.

This study has several limitations. The sample size is relatively small, and the within cohort analyses were exploratory, and so while our results present interesting future directions for research, higher-powered studies are needed to confirm these results. Additionally, cognitive and behavioural measures were not analysed as not all data could be obtained at the time of imaging. Incorporating such analyses could better contextualise the relationship between extra-motor tracts and changes in cognition and behaviour in ALS. Furthermore, including people living with ALS-FTD and bvFTD in future studies would provide a more comprehensive understanding of the white matter changes potentially underlying changes in appetitive behaviours, given the substantial variability in appetite disturbances across the disease spectrum. Finally, the limited number of acquired gradient directions and the limited resolution may contribute to partial volume effects and reduced angular resolution, which may limit our ability to resolve crossing fibres with low angular deltas.

Conclusions

This study reports fibre density and cross-section involvement of white matter tracts associated with appetite and behaviour in plwALS. Involvement of these tracts did not directly explain appetite and metabolic disturbances, which are likely shaped by a combination of grey matter alterations, neuroendocrine and peripheral influences, and psychosocial factors [2, 79]. We also demonstrate that fibre cross-section alterations of these non-motor tracts correlate with Δ FRS in plwALS, highlighting the interconnected nature of motor and non-motor systems. These findings add to growing evidence that ALS pathology is not confined to motor pathways but also involves circuits supporting reward, behaviour, and cognition, reinforcing its status as a multisystem neurodegenerative disease. Recognising this complexity is essential, as extra-motor involvement may contribute to the marked clinical heterogeneity observed in ALS. Future multimodal and longitudinal studies that integrate structural, functional, metabolic, and behavioural measures will be critical to disentangling these mechanisms and may ultimately inform the development of novel biomarkers and therapeutic strategies that address both motor and non-motor features of disease.

Abbreviations

ALS	Amyotrophic lateral sclerosis
plwALS	People living with amyotrophic lateral sclerosis
NND controls	Non-neurodegenerative disease controls
CNAQ	Council of Nutrition Appetite Questionnaire
ALSFRS-R	ALS Functional Rating Scale-Revised
Δ FRS	Change in Functional Rating Scale
FoV	Field of view
TR	Repetition time
TE	Echo time
T1w	T1 weighted
T2w	T2 weighted
fODF	Fibre orientation distribution function
FOD	Fibre orientation distributions
ANTs	Advanced Normalisation Tools
NODDI	Neurite orientation and dispersion density imaging
bvFTD	Behavioural-variant frontotemporal dementia
FWE	Family-wise error rate
FDR	False discovery rate

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12916-026-04763-6>.

Additional file 1: Table S1. Group differences in structural connectivity between Amyotrophic Lateral Sclerosis (N = 32) and non-neurodegenerative controls (N = 22). Table S2. Comparison of bilateral volumetric and cortical thickness measures between Amyotrophic Lateral Sclerosis (N = 32) and non-neurodegenerative controls (N = 22).

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Authors' contributions

FJS, STN, RDH, and PAM obtained funding for the study. FJS, STN, and JL designed the imaging study. FJS and STN managed the data collection and oversaw study ethics. FJS, STN, RDH, and PAM conducted data collection. TBS supervised the study. JC and HL conducted the formal analysis. JC wrote the manuscript. JC and TBS generated the figures. All authors contributed to the interpretation of the results; and read, revised, and approved the final manuscript.

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Data availability

The datasets generated and analysis code for the current study are available in the OpenNeuro repository, (<https://doi.org/10.18112/openneuro.ds005874.v1.1.0>). Supplementary video 1 is viewable on Figshare: (<https://doi.org/10.6084/m9.figshare.30511307>).

Declarations

Ethics approval and consent to participate

This study was approved by the University of Queensland, the Royal Brisbane and Women's Hospital (HREC/17/QRBW/616), and Uniting Care Health Human Research Ethics (Ref. No. 1801) Committees. The privacy rights of all participants were observed, and all participants provided written and informed consent to participate in the research and for deidentified data to be made available through publication and for other research purposes.

Consent for publication

Not applicable.

Competing interests

Frederik Steyn is a guest editor for the BMC Medicine special issue, Amyotrophic lateral sclerosis: latest advances and novel updates.

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